



Operating Instructions

Displacement sensor CD series

Laser type

- Thank you for purchasing. Check that the specifications agree with yours.
- Please read this manual through before using the sensor, and retain it for future reference.

1 Specifications

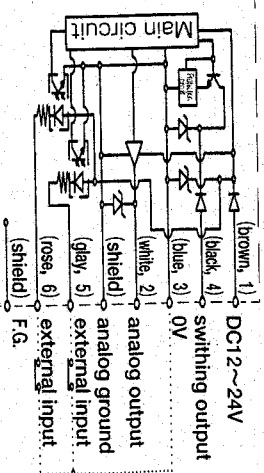
		Laser type				
Type	Cable type	CD3-30 (N,P)	CD3-50 (N,P)	CD3-80 (N,P)	CD3-100 (N,P)	CD3-250 (N,P)
	Connector type	CD3-30C (N,P)	CD3-50C (N,P)	CD3-80C (N,P)	CD3-100C (N,P)	CD3-250C (N,P)
Measuring range		30±4mm	50±10mm	80±15mm	100±40mm	250±150mm
Light source		Class 2 Red Laser diode (wave length 650nm max.1mW)				
Spot size (middle of range)		φ 0.5mm	φ 0.8mm	1×1.5mm	1×1.5mm	3×1.5mm
Supply voltage		DC12~24V (−5% +10%)				
Current consumption		max. 120mA/DC12V max. 80mA/DC24V Including analog output value				
Resolution 1)		4 μm	10 μm	10 μm	30 μm	150 μm
Linearity 2)		±1%FS				
Temperature drift		±0.08%FS/°C				
Response time		max.2.2ms (sensitivity: fix, averaging: 1) max.20.5ms (sensitivity: AUTO, averaging: 1)				
Outputs	Analog output	4~20mA				
	Switching output	NPN/PNP max.100mA/DC24V Residual voltage max.1.8V				
Indicators	Distance indicator	Near : Red				
		Middle : Orange				
		Far : Green				
		Out of range : Blinking Red/Green				
		Too high/low reflection : Blinking Red/Green				
	Stability indicator	Stable : Green				
		Unstable : Indicator OFF				
		Too high/low reflection : Red				
	Output indicator	ON status : Orange				
Ambient light		Sun light: max 10000 lx Incandescent lamp: max 3000 lx				
Operating temperature		−10~40°C				
Operating humidity		35~95%RH				
Housing material		Zinc die-cast				
Protection category		IP67				
Weight	Cable type	approx.130g (without cable)				
	Connector type	approx.140g				

FS (Full scale) is defined as, CD3-30:8mm, CD3-50:20mm, CD3-80:30mm, CD3-100:80mm, CD3-250:300mm

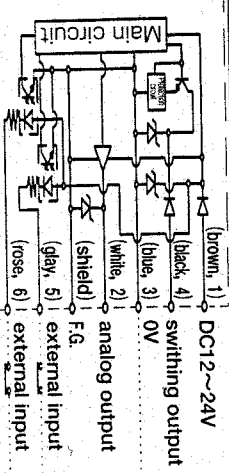
- 1) Middle of measuring range, Sensitivity: AUTO, Averaging: 64, Object: white ceramic
2) Linearity error (sensitivity: AUTO, averaging: 64, object: white ceramic)

2 Connection diagram

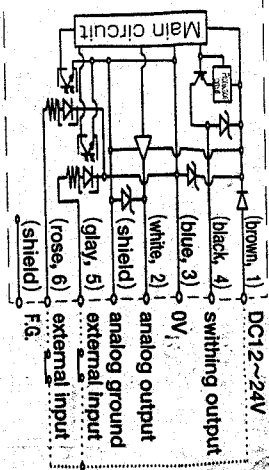
NPN Cable type



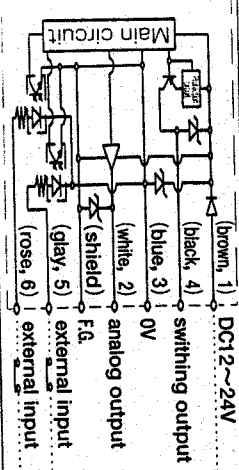
NPN Connector type



PNP Cable type

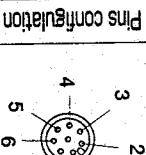


PNP Connector type



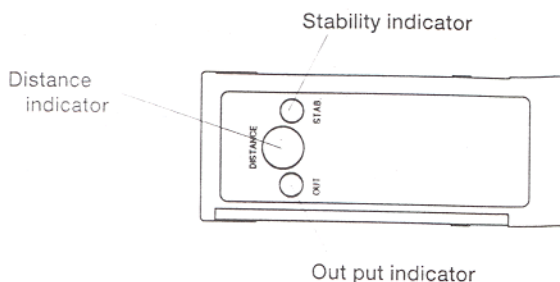
Caution for connection

- The F.G. shield wire of laser type is connected to the housing.
- Analog ground wire is not equipped for connector type. Therefore connect the analog ground terminal of analog input equipment and the 0V terminal of power supply.



- 1) Connect the lead wires correctly. The analog output wire must not be in contact with any other wire. Do not turn on any power while wiring.
 - 2) The blue wire (0V) and shield wire (analog GND) are internally connected.
- Use the blue wire (0V) for the power supply and use the shield wire (analog GND) for analog output.

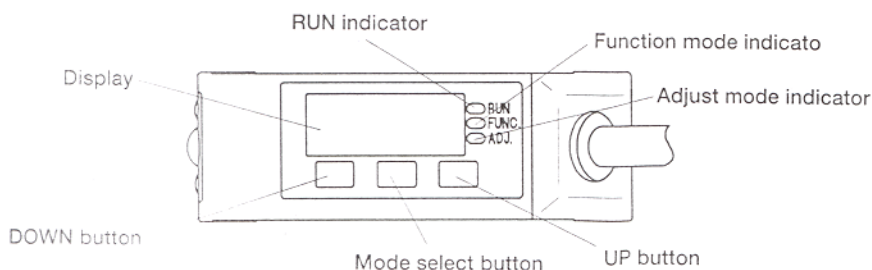
3] Functions of components



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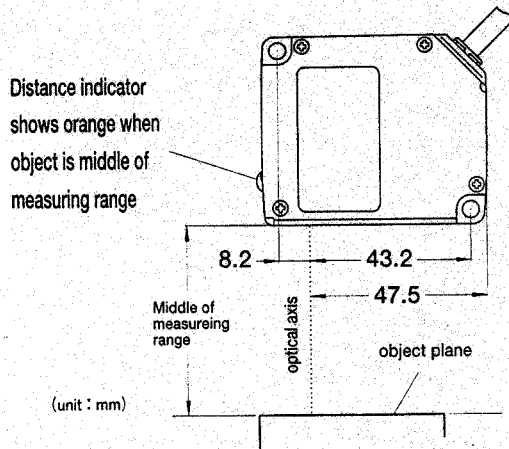
Distance indicator	This LED indicates the distance from sensor to object. <table border="1" style="margin-left: auto; margin-right: auto;"> <tr> <td>Blinking red and green</td><td>Red ON</td><td>Orange ON</td><td>Green ON</td><td>Blinking red and green</td></tr> </table> • This LED blink red and green when stability indicator is red ON due to too high/low reflection	Blinking red and green	Red ON	Orange ON	Green ON	Blinking red and green
Blinking red and green	Red ON	Orange ON	Green ON	Blinking red and green		
Stability indicator	This LED indicates the level of received light intensity. Green : Stable measuring OFF : Unstable measuring Red : Measurement is impossible due to too high/low level of Receiving light intensity.					
Output indicator	Orange : Switching output is ON status OFF : Switching output is OFF status					



RUN indicator	This LED indicates RUN mode.
Function mode indicator	This LED indicates function select mode.
Adjust mode indicator	This LED indicates adjust mode of function.
Display	RUN indicator ON : Indicate distance from sensor to object. Function mode indicator ON : Indicate function. Adjust mode indicator ON : Indicate value of selected function.
Mode select button	Mode is changed by this button.
UP/DOWN button	RUN mode : no effect Function select mode : Select function Adjust mode : Adjust value of function

4 INSTALLATION

Install the sensor and adjust the light spot onto the measuring point so that the distance indicator turns ON (orange) at the middle of measuring range.



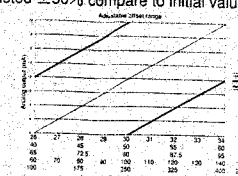
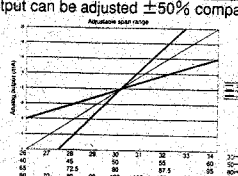
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- Adjust the sensor position so that the optical plane of the sensor is parallel with the plane of the object to be detected to obtain reliable measurement.
- In the laser type sensor, an invisible laser circle resides around the visible light spot. If there is any matter around the spot that is glossier than the measuring object it may lead to incorrect measurement.

				Adjust/Select Item	Teaching	Initial value																																													
Upper switching level	High	4.00...20.00 (mA) current indicate/indication off relative distance -9999...9999 (mm) 30mmtype, distance indicate -4.00...4.00 (mm) 50mmtype, distance indicate -10.00...10.00 (mm) 80mmtype, distance indicate -15.00...15.00 (mm) 100mmtype, distance indicate -40.00...40.00 (mm) 250mmtype, distance indicate -150.0...150.0 (mm)		Upper switching level can be adjusted without object.	Possible	Middle of measuring range +5%(FS)																																													
Lower switching level	Lo	same as above	same as above	Lower switching level can be adjusted without object.	Possible	Middle of measuring range -5%(FS)																																													
Output style (Timmer)	QtyP	only, oFdy, onEs, oFF		Timmer mode can be selected. only : on delay oFdy : off delay onEs : one shot oFF : without timmer		oFF																																													
Timmer period	tIEr	0...999, 1 S...10 S (ms)		Timmer period can be adjusted. 0...999ms: every 1ms 1...10S: every 1S		0																																													
Hysteresis	HySt	0.00...1.99 (mA) current indicate/indication off relative distance 0000...2499 (mm) 30mmtype, distance indicate 0.00...1.00 (mm) 50mmtype, distance indicate 0.00...2.49 (mm) 80mmtype, distance indicate 0.00...3.75 (mm) 100mmtype, distance indicate 0.00...10.00 (mm) 250mmtype, distance indicate 0.00...37.50 (mm)		Hysteresis of switching output can be adjusted.		0.15%(FS)																																													
Averaging	Arg	1, 4, 16, 64, 256, 1024		Averaging times of output can be selected.		64																																													
Sensitivity	SEnS	1...20 AUTO		Sensitivity can be selected. 1 (low)~20 (high) : Sensitivity can be fixed at selected sensitivity. AUTO : Sensitivity is adjusted automatically. Offset can be adjusted $\pm 50\%$ compare to initial value.	Possible	AUTO																																													
Offset	OFSt	-50.0P...50.0P			Possible	0.0P																																													
Span	SPAn	-50P...50P		Span of analog output can be adjusted $\pm 50\%$ compare to initial value. 		0P																																													
Analog output when reflection is too high/low	EoUt	Hold, CLp		Analog output can be selected when reflection is too high/low. Hold : Analog output is held just before reflection becomes too high/low. CLp : Analog output is fixed at 24mA.		CLp																																													
Display style during RUN status	dISp	4-20, dISt, 0-99, off, rdiS		Display style during RUN status can be selected. 4-20 : Current indicate (4-20mA) dISt : Distance indicate (mm) -4.000...4.000 30mm type -10.00...9.999...9.999, 10.00 50mm type -15.00...10.00...9.999...9.999, 10.00...15.00 80mm type -40.00...10.00...9.999...9.999, 10.00...40.00 100mm type -150.0...100.0...99.99...99.99, 100.0...150.0 250mm type 0-99 : Relative distance within range (-9999~9999) off : Display off rdiS : Reverse plus and minus when distance displayed.		dISt																																													
Bank	Ban	0, 1, 2, 3		4 pattern of function can be memorized (Except for external input customization and inside memory ON/OFF)		0																																													
External input customize	InPt	A Type A b Type B c Type C d Type D		Two external input (Rose wire and Grey wire) can be divided into four types. Type A (Bank select) <table border="1" data-bbox="892 1367 1237 1430"><tr><th>Bank No.</th><th>0</th><th>1</th><th>2</th><th>3</th></tr><tr><td>Rose wire</td><td>OFF</td><td>ON</td><td>OFF</td><td>ON</td></tr><tr><td>Gray wire</td><td>OFF</td><td>OFF</td><td>ON</td><td>ON</td></tr></table> Type B <table border="1" data-bbox="892 1430 1237 1473"><tr><td>Rose wire</td><td colspan="4">Laser off</td></tr><tr><td>Gray wire</td><td colspan="4">Reset</td></tr></table> Type C <table border="1" data-bbox="892 1473 1237 1517"><tr><td>Rose wire</td><td colspan="4">Sensitivity teaching</td></tr><tr><td>Gray wire</td><td colspan="4">Reset</td></tr></table> Type D <table border="1" data-bbox="892 1517 1237 1560"><tr><td>Rose wire</td><td colspan="4">Laser off</td></tr><tr><td>Gray wire</td><td colspan="4">Sensitivity teaching</td></tr></table> ON: Connect PNP to +V ON: Connect NPN to 0V	Bank No.	0	1	2	3	Rose wire	OFF	ON	OFF	ON	Gray wire	OFF	OFF	ON	ON	Rose wire	Laser off				Gray wire	Reset				Rose wire	Sensitivity teaching				Gray wire	Reset				Rose wire	Laser off				Gray wire	Sensitivity teaching					A
Bank No.	0	1	2	3																																															
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Interior memory ON/OFF	EEP	On, OFF		1. Choose OFF if you wish to keep operation without writing date of results into interior memory (Operation by external input only). 2. Choose OFF for the lifetime of interior memory (Limits: hundred thousand times). * 1. Change setting by button operation will be written on the memory. 2. Information which are not written on interior memory will be cleared when switched off.		oN																																													

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6 ADJUST / SELECT FUNCTION

① Press mode select button more than 3 seconds.
Then function is indicated. (function select mode)



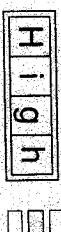
② Select function by pressing UP/DOWN button.



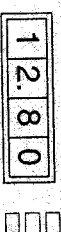
③ Press mode select button less than 3 seconds.
Then value of selected function is indicate. (adjust mode)



to ④



ex. Upper switching level is selected



④ Adjust/select value of function by pressing UP/DOWN button.



⑤ Press mode select button
more than 3 seconds.



Press mode select button
less than 3 seconds.

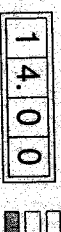


Back to RUN status.

Back to function select mode.
(another function can be adjusted)



ex. Adjusted at 14mA



● Reset of value of function

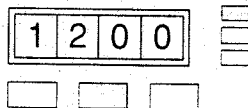
Turn on power supply with pressin mode select button, then every value of function is back tu initial status.

7 TEACHING

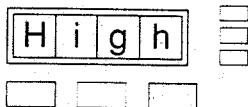
Switching level, sensitivity, offset are adjusted by teaching.

- ① Press mode select button more than 3 seconds.

Then function is indicated. (function select mode)



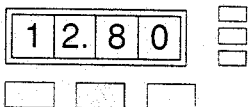
- ② Select function by pressing UP/DOWN button.



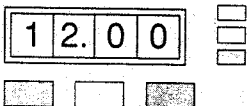
ex. Upper switching level is selected

- ③ Press mode select button less than 3 seconds.

Then value of selected function is indicated. (adjust mode)



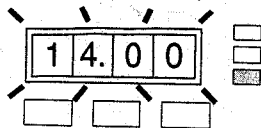
- ④ Place object at the adjust point, and press UP and DOWN button at the same time.



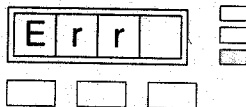
press same time

- ⑤ Blink adjusted value 3 time, then teaching is completed.

If error is indicated, confirm situation of installation.

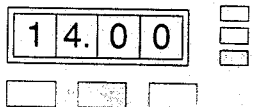


Blink 3 times



- ⑥ Press mode select button more than 3 seconds.

Press mode select button less than 3 seconds.



Back to RUN status.

Back to function select mode.
 (another function can be taught)

8 Warnings and cautions

Warnings

●Laser beam

○Laser models emit visible laser beam which complied with IEC825

Class 2. A warning and description label is attached to the side of the housing.

○DO NOT stare into the beam, or reflect the beam with a mirror.

○DO NOT disassemble the unit. Sensor are not equipped with auto-laser-off function.

Cautions

●DO NOT allow dust, oil, water, etc. to accumulate on the sensor face.

It degrades the sensor function. If pollution, just wipe clean using a dry cloth.

●When a switching regulator is to be used with a power supply, make sure to ground the frame ground terminal.

●DO NOT use the sensor in a transient state at power on (Approx. 30min. warm-up time).

●DO NOT run the sensor cable near a high-voltage lines, or power lines or put them together in the same raceway. This warning should be strictly observed to prevent malfunctions caused by inductive interference.

●Use of controls or adjustments or performance of procedures other than those specified herein may result in hazardous radiation exposure.

●Sensors equipped on machinery are under FDA of American laser regulation. CD3 series have already been registered with CDRH (Center for Devices and Radiological Health).

Only
connector type



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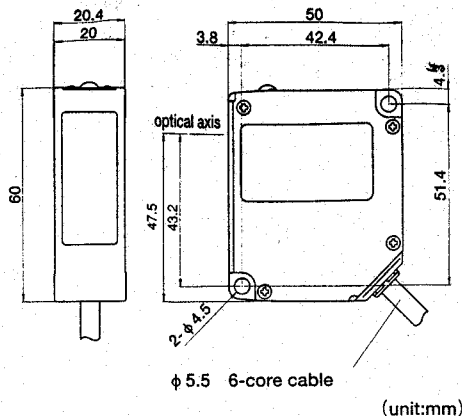
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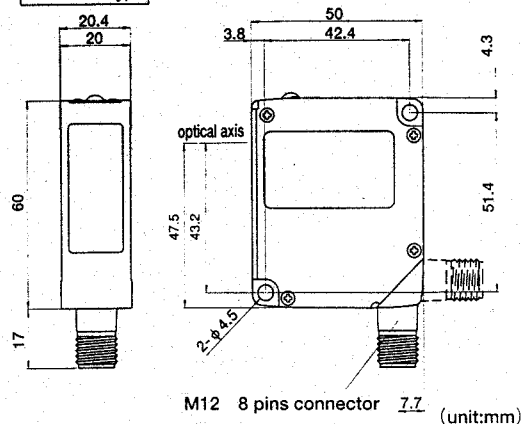
THIS PRODUCT IS NOT DESIGNED TO BE USED AS A SAFETY DEVICE TO PROTECT HUMAN BODY.

9 Dimension drawing

Cable type



Connector type



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